

Do Platform-Exclusive Games Reduce Rival Gaming App Usage?

The Case of Anipang on Kakao

Marketing Models — Assignment 2

Group 11

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Research Question & Framework

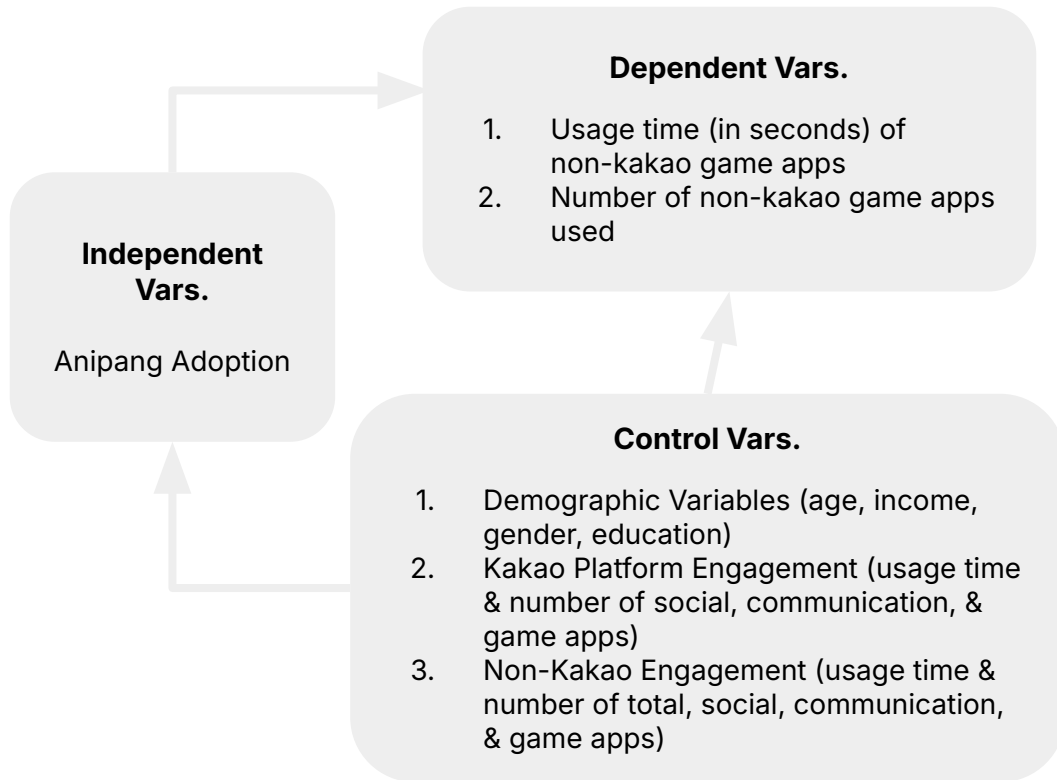
Business Question:

Provide useful & effective strategies for mobile app platform owners & app developers after a popular app has been released on a platform

Research Question:

*Does the adoption of a popular platform-exclusive game (Anipang) lead to a **decline in the usage of rival non-platform mobile games**?*

Conceptual Framework



Data Overview

Dataset	kakao_all.csv
Data Type	Individual-level weekly panel data on app usage count and time (Android devices)
Users (N)	849 Android mobile users
Time Period (T)	2 weeks (July 23 – August 5, 2012): • Week 1: Pre-Adoption of Anipang • Week 2: Post-Adoption of Anipang

Key Variables

User, Time, Treatment Variables

Demographic Profile Variables

Variable	Description	Variable	Description
panel_id	User ID	age	1: 7 ~ 18 2: 19 ~ 29 3: 30 ~ 39 4: 40 ~ 49 5: 50 ~ 69
week	1: before the release of Anipang (Jul 23-29, 2012) 2: after the release of Anipang (Jul 30-Aug 5, 2012)	income	1: < \$1,000 2: \$1,000 ~ \$3,000 3: \$3,000 ~ \$5,000 4: > \$5,000
tg	1: treatment group (Anipang adopters) 0: control group (Anipang non-adopters)	education	1: Elementary, middle, and high school students 2: High school graduates 3: Undergraduate or graduate students 4: College graduates
ii	1: if a user adopted Anipang at or before the current week; 0: otherwise	gender	1: Male, 0: Female

Key Variables

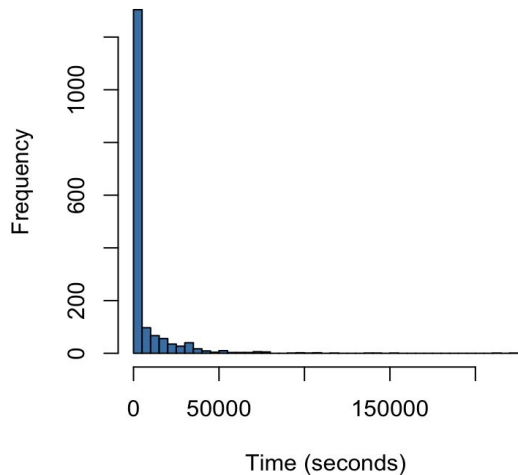
App Usage Within Kakao Platform

App Usage Outside Kakao Platform

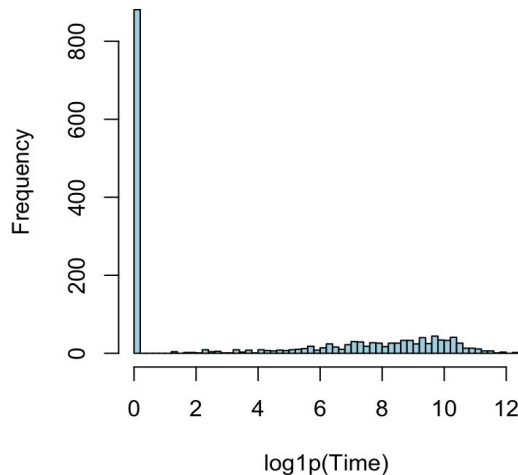
Variable	Description	Variable	Description
log_t_kakao_talk	Log-transformed usage time of Kakao Talk (in seconds)	log_t_non_kakao_talk	Log-transformed usage time of communication apps (in seconds)
log_t_kakao_story	Log-transformed usage time of Kakao Story (excluding Anipang) (in seconds)	log_t_non_kakao_story	Log-transformed usage time of social networking apps (in seconds)
log_t_kakao_game	Log-transformed usage time of Kakao Game apps (in seconds)	log_t_non_kakao_game	Log-transformed usage time of game apps (in seconds)
log_t_anipang	Log-transformed usage time of Anipang (in seconds)	log_t_non_kakao	Log-transformed usage time of all apps (in seconds)
n_kakao_game	Number of Kakao Game apps used (excluding Anipang)	n_non_kakao_talk	Number of communication apps used
		n_non_kakao_story	Number of social networking apps used
		n_non_kakao_game	Number of game apps used
		n_non_kakao	Number of all apps used

Log-Transforming Usage Time of Non-Kakao Games for Improved Distribution

Original t_non_kakao_game



Log-Transformed t_non_kakao_game



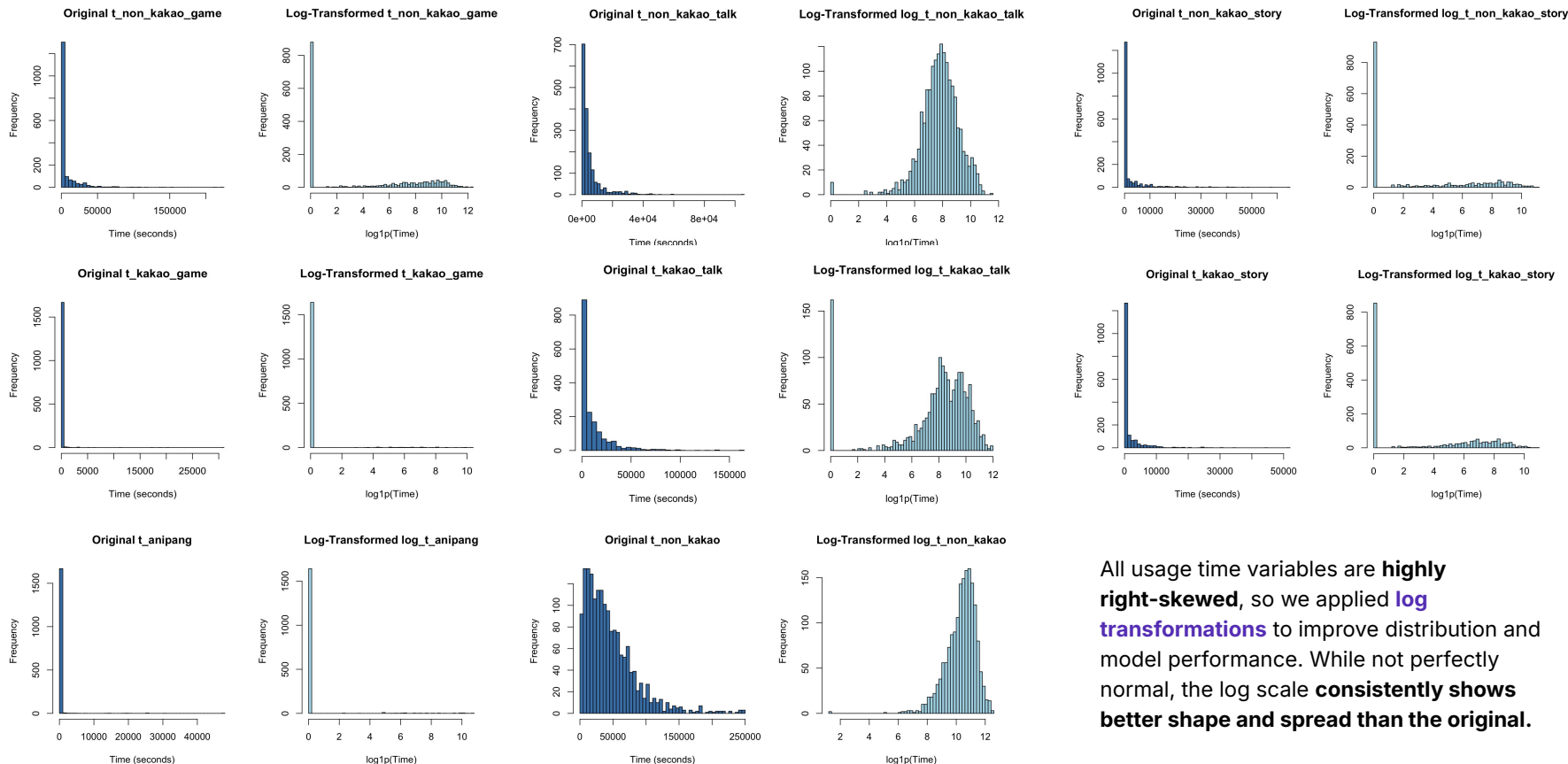
Usage time of rival (non-Kakao) games highly **right-skewed**, with median of 0 & extreme outliers

Log transformation reduces skewness and compresses the scale, improving distribution shape and spread

Min	Q1	Med	Mean	Q3	Max
0	0	0	6200	3976	221594

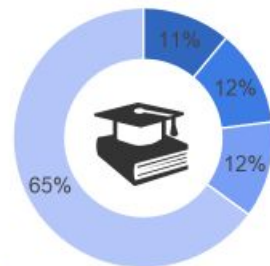
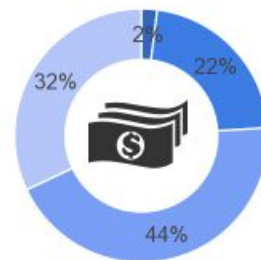
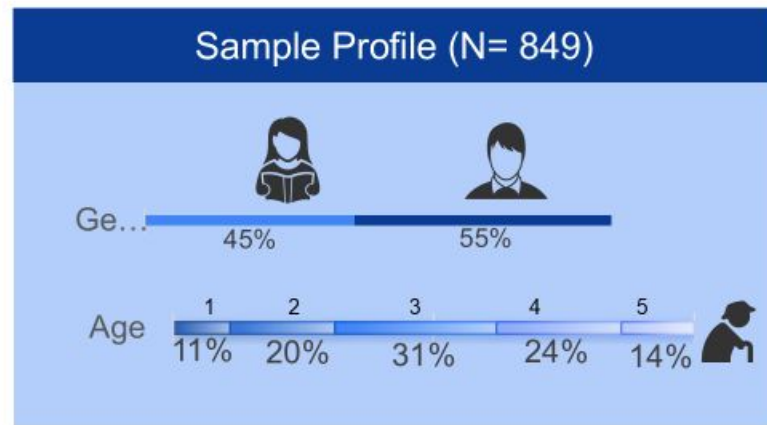
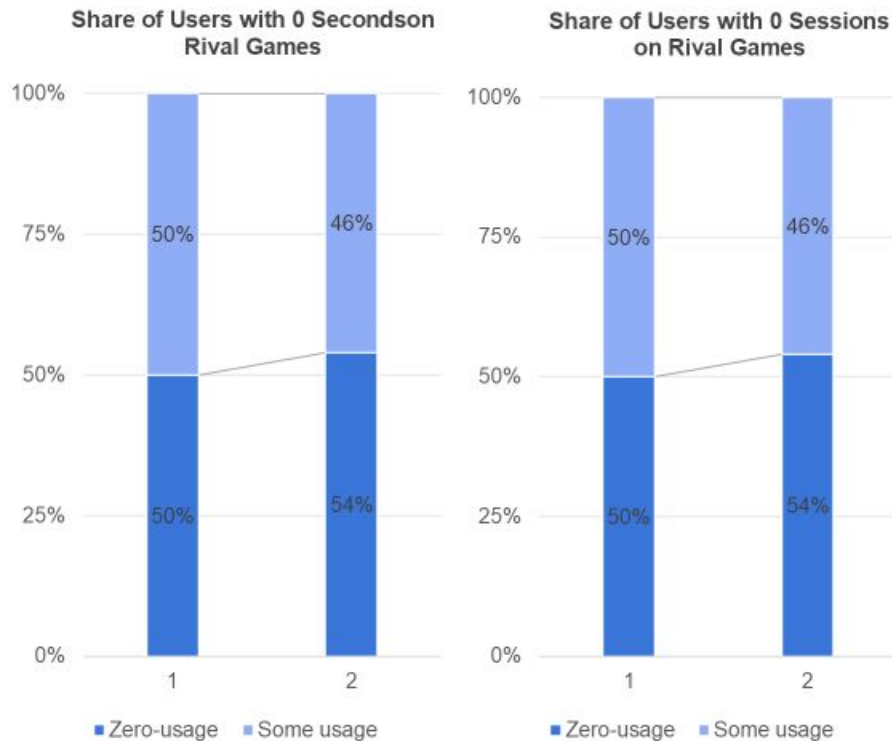
Min	Q1	Med	Mean	Q3	Max
0	0	0	3.582	8.288	12.309

Log-Transforming All Usage Time Variables for Improved Distribution



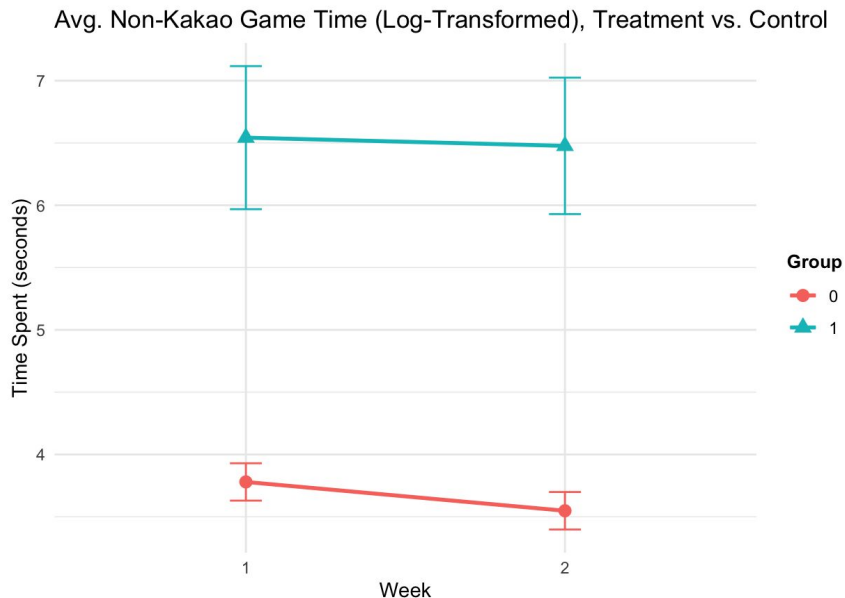
All usage time variables are **highly right-skewed**, so we applied **log transformations** to improve distribution and model performance. While not perfectly normal, the log scale **consistently shows better shape and spread than the original**.

Descriptive Statistics



■ 1 ■ 2 ■ 3 ■ 4

Preliminary Analysis: Rival (Non-Kakao) Game Usage Time by Treatment Group

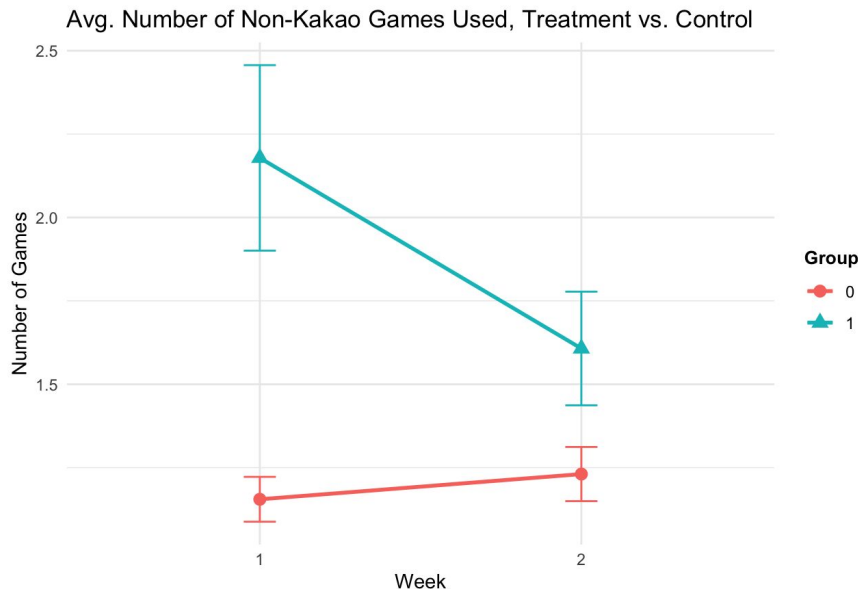


0: Control Group
(People who never downloaded Anipang)

1: Treatment Group
(People who downloaded Anipang)

- **Treatment group:** Log-transformed average game time decreased minimally from 6.54 (pre-adoption) to 6.48 seconds (post-adoption)
- **Control group:** Decreased slightly from 3.78 to 3.55 seconds over same time period
- Visually, there seems to be **no noticeable behavioral change in rival game usage within the treatment group after Anipang adoption**, but formal testing is needed to assess statistical significance

Preliminary Analysis: Number of Rival (Non-Kakao) Games Used by Treatment Group



0: Control Group
(People who never downloaded Anipang)

1: Treatment Group
(People who downloaded Anipang)

- **Treatment group:** Used an average of 2.18 non-Kakao games in Week 1 (pre-adoption), dropping to 1.61 in Week 2 (post-adoption)
- **Control group:** Increased slightly from 1.16 to 1.23 over the same time period
- The **treatment group appears to reduce number of non-Kakao game use post-adoption**, but formal testing is needed to assess statistical significance

PSM: Selecting Covariates to Reduce Selection Bias

Type	Variables	Rationale
Demographic Controls	age	- Age, gender, education, & income often influence mobile game adoption due to differences in tech familiarity, preferences, & access - Including these ensures balanced comparison groups based on key user characteristics
	income	
	education	
	gender	
Controls for Existing Kakao Platform / Kakao Game Interest	log_t_kakao_game	- Users already active on Kakao Platform &/or Kakao Games are more likely to adopt Anipang - Matching on this helps separate the effect of Anipang adoption from pre-existing preferences for Kakao Platform / Kakao Games
	n_kakao_game	
	log_t_kakao_talk	
	log_t_kakao_story	
Controls for Total Non-Kakao Mobile Activity	log_t_non_kakao	- Captures general app usage habits beyond Kakao - Helps ensure post-adoption differences aren't simply due to one group being more or less active on mobile overall
	n_non_kakao	
	log_t_non_kakao_talk	
	n_non_kakao_talk	
	log_t_non_kakao_story	
	n_non_kakao_story	
Controls for General Gaming Behavior Outside the Platform	log_t_non_kakao_game	Ensures adopters and non-adopters had similar pre-existing levels of mobile gaming behavior outside Kakao
	n_non_kakao_game	

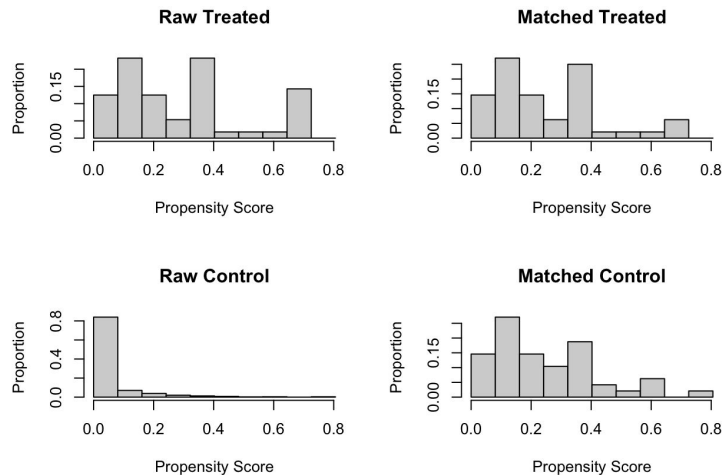
PSM: Overview of 10 Matching Strategies

	Matching	Replacement	Caliper
PSM 1	1:1	Without	Without
PSM 2	1:1	Without	0.05
PSM 3	1:1	Without	0.10
PSM 4	1:1	Without	0.25
PSM 5	2:1	Without	Without
PSM 6	2:1	Without	0.1
PSM 7	2:1	Without	0.05
PSM 8	2:1	With	Without
PSM 9	3:1	Without	Without
PSM 10	3:1	Without	0.05

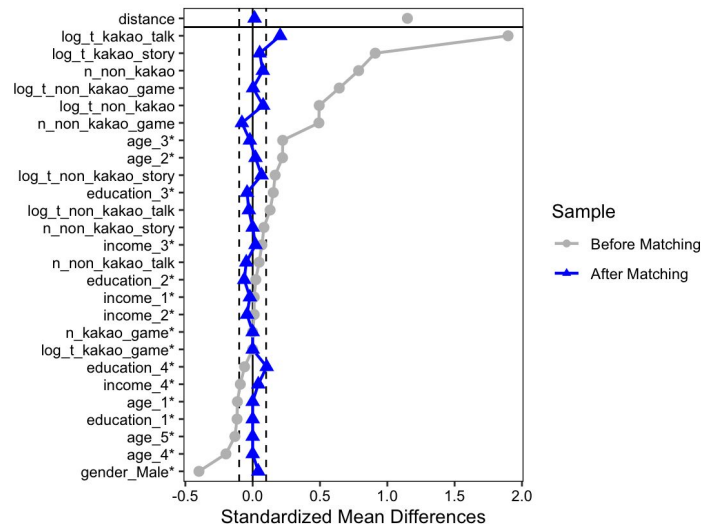
While all 10 PSMs improved covariate balance over the original data, **we focus on matching results with 1:1 matching (no replacement, 0.25 caliper)**, which yielded the most consistent balance across covariates for subsequent DID analysis.

PSM Matching Outcomes: Improved Balance & Reduced Selection Bias

1:1 Matching, No Replacement, 0.25 Caliper



Standardized Mean Differences Before and After Matching



	Control	Treated
All	793	56
Matched	48	48
Unmatched	748	8
Discarded	0	0

Post-Matching:

- **Propensity scores of treated & control groups are more aligned**, improving validity of causal inference
- **Standardized mean differences across covariates are notably closer to 0**, indicating better balance

DID: Selecting Control Variables to Reduce Omitted Variable Bias

$$Y_{it} = \beta_0 + \beta_1 ii_{it} + \gamma X_{it} + \alpha_i + \delta_t + u_{it}$$

- **Y_{it}** = log time or session count on rival (non-platform) games
- **ii_{it}** = 1 if user *i* adopted Anipang in Week 2, 0 otherwise
- **X_{it}** = time-varying controls (see under)
- **α_i** = user fixed effects
- **δ_t** = week fixed effects
- **u_{it}** = random error

Key parameter:

β₁ = average change in rival-game usage attributable to Anipang adoption

(Positive = crowd-in, Negative = crowd-out)

Null hypothesis (H₀): β₁ = 0 → Anipang adoption has **no effect** on rival-game usage

Platform engagement	Cross-platform behaviour
log t_kakao_game & n_kakao_game	log t_non_kakao & n_non_kakao
log t_kakao_talk	log t_non_kakao_talk & n_non_kakao_talk
log t_kakao_story	log t_non_kakao_story & n_non_kakao_story

*Static demographics (gender, income, education, baseline age code) are absorbed by **α_i** and therefore not entered separately.

Evaluating Anipang Adoption Effects on Time Usage of Rival Games: Dummy & FE DID Models

Daniel

```
log_t_non_kakao_game ~ ii + panel_id + week + log_t_kakao_talk + log_t_kakao_story +  
log_t_non_kakao_story + log_t_kakao_game + log_t_non_kakao_talk + log_t_non_kakao +  
n_non_kakao_talk + n_non_kakao_story + n_kakao_game + n_non_kakao
```

Dummy DID Coefficients	Estimate	Std. Error	z value	Pr(> z)
Intercept	7.584	5.666	1.338	0.184
ii1	0.269	0.551	0.488	0.627
panel_id26922	-8.577	2.063	-4.158	~0 ***
week2	-0.381	0.326	-1.169	0.246
log_t_non_kakao_story	0.292	0.186	1.568	0.121
log_t_non_kakao_talk	-0.039	0.275	-0.141	0.888
log_t_non_kakao	0.643	0.511	1.258	0.212
n_non_kakao_talk	-0.515	0.269	-1.914	0.059 .
n_non_kakao_story	-0.566	0.738	-0.768	0.445
n_non_kakao_game	0.862	0.172	5.001	~0 ***
n_non_kakao	-0.038	0.042	-0.911	0.365

*** significant at 0.1% level; ** significant at 1% level, * significant at 5% level

**not displaying all
model outputs*

Multiple R-squared: 0.939

Overall Model Significance: p-value ~ 0

- **ii1 coefficient:** Adopting Anipang (ii1=1) is associated with a 0.269 increase in log-time spent on rival (non-Kakao) games, compared to non-adopters (ii1 = 0)
- **Not significant:** However, this effect isn't statistically significant ($p = 0.627$), so we fail to reject the null hypothesis
- **RQ implication:** No evidence that adopting Anipang leads to a measurable change in time spent on rival non-Kakao games

Evaluating Anipang Adoption Effects on Time Usage of Rival Games: Dummy & FE DID Models

- **ii1 coefficient:** Adopting Anipang (ii1=1) is associated with a 0.126 increase in log-time spent on rival (non-Kakao) games, compared to non-adopters (ii1 = 0)
- **Not significant:** However, this effect isn't statistically significant ($p = 0.841$), so we fail to reject the null hypothesis
- **RQ implication:** No evidence that adopting Anipang leads to a measurable change in time spent on rival non-Kakao games

```
log_t_non_kakao_game ~ ii + week + log_t_kakao_game + n_kakao_game + log_t_non_kakao
+ n_non_kakao + log_t_kakao_talk + log_t_kakao_story + log_t_non_kakao_talk +
n_non_kakao_talk + log_t_non_kakao_story + n_non_kakao_story
```

FE Estimation Coefficients	Estimate	Std. Error	z value	Pr(> z)
ii1	0.126	0.624	0.202	0.841
week2	-0.427	0.369	-1.155	0.251
log_t_kakao_game	0.19	0.154	1.232	0.221
n_kakao_game	-0.493	0.511	-0.965	0.337
log_t_non_kakao	0.404	0.577	0.7	0.486
n_non_kakao	0.053	0.043	1.243	0.218
log_t_kakao_talk	-0.373	0.314	-1.189	0.238
log_t_kakao_story	-0.2	0.149	-1.347	0.182
log_t_non_kakao_talk	-0.108	0.312	-0.346	0.73
n_non_kakao_talk	-0.725	0.302	-2.406	0.018 *
log_t_non_kakao_story	0.315	0.211	1.494	0.139
n_non_kakao_story	-1.067	0.829	-1.288	0.201

*** significant at 0.1% level; ** significant at 1% level, * significant at 5% level

R-squared: 0.158

Overall Model Significance: p-value = 0.226

Evaluating Anipang Adoption Effects on Time Usage of Rival Games: Dummy & FE DID Models

`log_t_non_kakao_game ~ ii + panel_id + week + log_t_kakao_talk + log_t_kakao_story + log_t_non_kakao_story + log_t_kakao_game + log_t_non_kakao_talk + log_t_non_kakao + n_non_kakao_talk + n_non_kakao_story + n_kakao_game + n_non_kakao`

Dummy DID Coefficients	Estimate	Std. Error	z value	Pr(> z)
Intercept	7.584	5.666	1.338	0.184
ii1	0.269	0.551	0.488	0.627
panel_id26922	-8.577	2.063	-4.158	~0 ***
week2	-0.381	0.326	-1.169	0.246
log_t_non_kakao_story	0.292	0.186	1.568	0.121
log_t_non_kakao_talk	-0.039	0.275	-0.141	0.888
log_t_non_kakao	0.643	0.511	1.258	0.212
n_non_kakao_talk	-0.515	0.269	-1.914	0.059 .
n_non_kakao_story	-0.566	0.738	-0.768	0.445
n_non_kakao_game	0.862	0.172	5.001	~0 ***
n_non_kakao	-0.038	0.042	-0.911	0.365

`log_t_non_kakao_game ~ ii + week + log_t_kakao_game + n_kakao_game + log_t_non_kakao + n_non_kakao + log_t_kakao_talk + log_t_kakao_story + log_t_non_kakao_talk + n_non_kakao_talk + log_t_non_kakao_story + n_non_kakao_story`

FE Estimation Coefficients	Estimate	Std. Error	z value	Pr(> z)
ii1	0.126	0.624	0.202	0.841
week2	-0.427	0.369	-1.155	0.251
log_t_kakao_game	0.19	0.154	1.232	0.221
n_kakao_game	-0.493	0.511	-0.965	0.337
log_t_non_kakao	0.404	0.577	0.7	0.486
n_non_kakao	0.053	0.043	1.243	0.218
log_t_kakao_talk	-0.373	0.314	-1.189	0.238
log_t_kakao_story	-0.2	0.149	-1.347	0.182
log_t_non_kakao_talk	-0.108	0.312	-0.346	0.73
n_non_kakao_talk	-0.725	0.302	-2.406	0.018 *
log_t_non_kakao_story	0.315	0.211	1.494	0.139
n_non_kakao_story	-1.067	0.829	-1.288	0.201

Across both the Dummy & FE DID models, the coefficient for ii1 is **consistently positive but not statistically significant**, suggesting no clear evidence that Anipang adoption affects time spent on rival (non-Kakao) games

Evaluating Anipang Adoption Effects on Number of Rival Games Used: Dummy vs. FE DID Models

Ernest

```
n_non_kakao_game ~ ii + panel_id + week + log_t_kakao_talk + log_t_kakao_story +  
log_t_non_kakao_story + log_t_kakao_game + log_t_non_kakao_talk + log_t_non_kakao +  
n_non_kakao_talk + n_non_kakao_story + n_kakao_game + n_non_kakao
```

Dummy DID Coefficients	Estimate	Std. Error	z value	Pr(> z)
Intercept	2.054	3.19	0.644	0.522
ii1	-0.2	0.308	-0.65	0.517
panel_id26922	-0.068	1.266	-0.054	0.957
week2	0.062	0.183	0.336	0.737
log_t_non_kakao_story	-0.058	0.105	-0.548	0.585
log_t_non_kakao_talk	-0.051	0.154	-0.333	0.74
log_t_non_kakao	-0.386	0.285	-1.354	0.179
n_non_kakao_talk	-0.049	0.154	-0.318	0.752
n_non_kakao_story	-0.295	0.412	-0.715	0.476
n_non_kakao_game	-0.881	0.253	-3.484	0.001 ***
n_non_kakao	0.092	0.021	4.313	0 ***

*** significant at 0.1% level; ** significant at 1% level, * significant at 5% level

**not displaying all
model outputs*

Multiple R-squared: 0.924

Overall Model Significance: p-value ~ 0

- **ii1 coefficient:** Adopting Anipang (ii1 = 1) is associated with a 0.2 decrease in the number of rival (non-Kakao) games used, compared to non-adopters (ii1 = 0)
- **Not significant:** However, this effect isn't statistically significant ($p = 0.517$), so we fail to reject the null hypothesis
- **RQ implication:** No evidence that Anipang adoption leads to a measurable change in the number of rival non-Kakao games used

Evaluating Anipang Adoption Effects on Number of Rival Games Used: Dummy vs. FE DID Models

- **ii1 coefficient:** Adopting Anipang (ii1 = 1) is associated with a 0.166 decrease in the number of rival (non-Kakao) games used, compared to non-adopters (ii1 = 0)
- **Not significant:** However, this effect isn't statistically significant ($p = 0.635$), so we fail to reject the null hypothesis
- **RQ implication:** No evidence that Anipang adoption leads to a measurable change in the number of rival non-Kakao games used

```
n_non_kakao_game ~ ii + week + log_t_kakao_game + n_kakao_game + log_t_non_kakao +
n_non_kakao + log_t_kakao_talk + log_t_kakao_story + log_t_non_kakao_talk + n_non_kakao_talk
+ log_t_non_kakao_story + n_non_kakao_story
```

FE Estimation Coefficients	Estimate	Std. Error	z value	Pr(> z)
ii1	-0.166	0.349	-0.476	0.635
week2	-0.053	0.206	-0.257	0.798
log_t_kakao_game	0.189	0.086	2.202	0.03 *
n_kakao_game	-1.014	0.285	-3.554	0.001 ***
log_t_non_kakao	-0.277	0.322	-0.861	0.392
n_non_kakao	0.106	0.024	4.436	0 ***
log_t_kakao_talk	-0.069	0.175	-0.395	0.694
log_t_kakao_story	-0.07	0.083	-0.842	0.402
log_t_non_kakao_talk	-0.08	0.174	-0.46	0.646
n_non_kakao_talk	-0.244	0.168	-1.447	0.152
log_t_non_kakao_story	0.027	0.118	0.229	0.819
n_non_kakao_story	-0.582	0.463	-1.257	0.212

*** significant at 0.1% level; ** significant at 1% level, * significant at 5% level

Multiple R-squared: 0.334
Overall Model Significance: p-value = 0.003

Evaluating Anipang Adoption Effects on Number of Rival Games Used: Dummy vs. FE DID Models

`n_non_kakao_game ~ ii + panel_id + week + log_t_kakao_talk + log_t_kakao_story + log_t_non_kakao_story + log_t_kakao_game + log_t_non_kakao_talk + log_t_non_kakao + n_non_kakao_talk + n_non_kakao_story + n_kakao_game + n_non_kakao`

Dummy DID Coefficients	Estimate	Std. Error	t value	Pr(> t)
Intercept	2.054	3.19	0.644	0.522
ii1	-0.2	0.308	-0.65	0.517
panel_id26922	-0.068	1.266	-0.054	0.957
week2	0.062	0.183	0.336	0.737
log_t_non_kakao_story	-0.058	0.105	-0.548	0.585
log_t_non_kakao_talk	-0.051	0.154	-0.333	0.74
log_t_non_kakao	-0.386	0.285	-1.354	0.179
n_non_kakao_talk	-0.049	0.154	-0.318	0.752
n_non_kakao_story	-0.295	0.412	-0.715	0.476
n_non_kakao_game	-0.881	0.253	-3.484	0.001 ***
n_non_kakao	0.092	0.021	4.313	0 ***

`n_non_kakao_game ~ ii + week + log_t_kakao_game + n_kakao_game + log_t_non_kakao + n_non_kakao + log_t_kakao_talk + log_t_kakao_story + log_t_non_kakao_talk + n_non_kakao_talk + log_t_non_kakao_story + n_non_kakao_story`

FE Estimation Coefficients	Estimate	Std. Error	z value	Pr(> z)
ii1	-0.166	0.349	-0.476	0.635
week2	-0.053	0.206	-0.257	0.798
log_t_kakao_game	0.189	0.086	2.202	0.03 *
n_kakao_game	-1.014	0.285	-3.554	0.001 ***
log_t_non_kakao	-0.277	0.322	-0.861	0.392
n_non_kakao	0.106	0.024	4.436	0 ***
log_t_kakao_talk	-0.069	0.175	-0.395	0.694
log_t_kakao_story	-0.07	0.083	-0.842	0.402
log_t_non_kakao_talk	-0.08	0.174	-0.46	0.646
n_non_kakao_talk	-0.244	0.168	-1.447	0.152
log_t_non_kakao_story	0.027	0.118	0.229	0.819
n_non_kakao_story	-0.582	0.463	-1.257	0.212

Across both Dummy & FE DID models, the **ii1** coefficient is **consistently negative but not statistically significant** — suggesting no clear evidence that Anipang adoption affects usage of rival non-Kakao games

Summary of Results: Log-Transformed Time Usage (in Seconds) of Non-Kakao (Rival) Games

Dummy DID

	ii		ii
Model 1	0.559 (NS)	Model 6	0.699 (NS)
Model 2	-0.146 (NS)	Model 7	0.967 (NS)
Model 3	-0.144 (NS)	Model 8	0.629 (NS)
Model 4	0.269 (NS)	Model 9	0.660 (NS)
Model 5	0.566 (NS)	Model 10	0.300 (NS)

FE Estimation

	ii		ii
Model 1	0.265 (NS)	Model 6	0.567 (NS)
Model 2	-0.497 (NS)	Model 7	0.457 (NS)
Model 3	-0.429 (NS)	Model 8	N/A (W/ Replacement)
Model 4	0.126 (NS)	Model 9	0.058 (NS)
Model 5	0.986 (NS)	Model 10	0.062 (NS)

Across all 10 models:

- The coefficient for ii (Anipang adoption) is not statistically significant
- The direction of the effect (positive or negative) varies, but none provide consistent or reliable evidence
- **Conclusion:** There is **no clear support** that adopting Anipang impacts **time spent on rival non-Kakao games**

Summary of Results: Number of Non-Kakao (Rival) Games

Dummy DID

	ii		ii
Model 1	-0.506 (NS)	Model 6	0.058 .
Model 2	-0.295 (NS)	Model 7	0.253 (NS)
Model 3	-0.259 (NS)	Model 8	-0.753 (NS)
Model 4	-0.200 (NS)	Model 9	-0.540 (NS)
Model 5	0.175 (NS)	Model 10	-0.213 (NS)

FE Estimation

	ii		ii
Model 1	-0.435 (NS)	Model 6	0.105 .
Model 2	-0.410 (NS)	Model 7	0.126 (NS)
Model 3	-0.358 (NS)	Model 8	N/A (W/ Replacement)
Model 4	-0.166 (NS)	Model 9	-0.508 (NS)
Model 5	0.091 (NS)	Model 10	-0.074 (NS)

Across all 10 models:

- The coefficient for ii (Anipang adoption) is not statistically significant
- The direction of the effect (positive or negative) varies, but none provide consistent or reliable evidence
- **Conclusion:** There is **no clear support** that adopting Anipang impacts **number of rival non-Kakao games used**

Conclusion

Answering the Research Question:

RQ: Does the adoption of a popular platform-exclusive game (Anipang) lead to a decline in the usage of rival non-platform mobile games?

- Anipang adoption **did not** significantly decrease the usage time or number of rival (non-Kakao) gaming apps
- Launching a popular exclusive game like Anipang **does not** cannibalize time from rival platforms
- **Consistent insignificant results** across 10 matched DID models

Marketing and Business Implications

BQ: Provide useful & effective strategies for mobile app platform owners & app developers after a popular app has been released on a platform.

For Developers

A new platform-exclusive hit doesn't pose an immediate threat to user retention.

- Instead of rushing to react, developers can stay the course, as a viral platform-exclusive game is unlikely to cause a sudden user drop-off
- Instead of replicating the viral game, developers should focus on developing its own niche title, as the new hit won't displace existing games.

For Platform (e.g. Kakao)

New viral platform-based games don't displace existing non-platform habitual gaming; they likely add additional playing time on top of it.

- Viral hits like Anipang may attract users, but platforms should not assume they boost total engagement without further evidence
- Focus on increasing engagement with the platform internally, rather than trying to cannibalize other platforms

**Additional research on how new viral platform-based games affect within-Platform engagement is needed for further implications*

Limitations

Short observation window (2 weeks)



Single hit & single platform, cannot generalize to other social platforms



Only usage metrics (time / sessions)



Observational PSM + DID—possible hidden bias



Study focuses solely on effect of within-platform game adoption on outside-platform engagement



Future Research

Track adopters for several months to detect any delayed cannibalisation or churn

Replicate with other blockbuster apps and on multiple app stores to test generalisability

Add spending & ad-revenue data to gauge monetization effects

Use natural experiments or platform-run A/B tests to strengthen causal claims

Expand research scope beyond outside-platform engagement (e.g. user satisfaction)

Thank You!

Questions or Feedback?

Appendix: R Codes

#Read in dataset, factorize & log-transform variables

```
library(dplyr)
library(MatchIt)
library(ggplot2)
setwd("~/Applications/IMC462/HW2")
kakao = read.csv("kakao_all.csv")

# Factorize vars
kakao$tg = as.factor(kakao$tg)
kakao$age <- as.factor(kakao$age)
kakao$ii <- as.factor(kakao$ii)
kakao$income <- as.factor(kakao$income)
kakao$gender <- as.factor(kakao$gender)
kakao$education <- as.factor(kakao$education)
kakao$week <- as.factor(kakao$week)
kakao$panel_id <- as.factor(kakao$panel_id)

# Add log-transformed variables using log1p()
kakao$log_t_kakao_game <- log1p(kakao$t_kakao_game)
kakao$log_t_non_kakao_game <- log1p(kakao$t_non_kakao_game)
kakao$log_t_non_kakao <- log1p(kakao$t_non_kakao)
kakao$log_t_kakao_talk <- log1p(kakao$t_kakao_talk)
kakao$log_t_non_kakao_talk <- log1p(kakao$t_non_kakao_talk)
kakao$log_t_kakao_story <- log1p(kakao$t_kakao_story)
kakao$log_t_non_kakao_story <- log1p(kakao$t_non_kakao_story)
kakao$log_t_anipang <- log1p(kakao$t_anipang)
```

#Descriptive Statistics

```
# Side by Side histogram
par(mfrow = c(1, 2))
```

```
# Original
hist(kakao$t_non_kakao,
     breaks = 50,
     col = "steelblue",
     main = "Original t_non_kakao",
     xlab = "Time (seconds)",
     ylab = "Frequency")
```

```
# Log-Transformed
hist(kakao$log_t_non_kakao,
     breaks = 50,
     col = "lightblue",
     main = "Log-Transformed log_t_non_kakao",
     xlab = "log1p(Time)",
     ylab = "Frequency")
```

```
#time usage of non kakao games: treatment vs. control
# Summarize average non-Kakao game time by week and treatment group
summary_plot_data <- kakao %>%
  group_by(week, tg) %>%
  summarise(
    mean_time = mean(log_t_non_kakao_game, na.rm = TRUE),
    se = sd(log_t_non_kakao_game, na.rm = TRUE) / sqrt(n()),
    .groups = "drop"
  )
```

```
# Plot
ggplot(summary_plot_data, aes(x = week, y = mean_time, group = tg, color = tg,
  shape = tg)) +
  geom_line(size = 1) +
  geom_point(size = 3) +
  geom_errorbar(aes(ymin = mean_time - se, ymax = mean_time + se), width = 0.1)
+
  scale_shape_manual(values = c("0" = 16, "1" = 17)) + # 16 = circle (control), 17 =
triangle (treatment)
  labs(
    title = "Avg. Non-Kakao Game Time (Log-Transformed), Treatment vs. Control",
    x = "Week",
    y = "Time Spent (seconds)",
    color = "Group",
    shape = "Group"
  ) +
  theme_minimal() +
  theme(
    legend.position = "right",
    legend.title = element_text(size = 10, face = "bold"),
    legend.text = element_text(size = 9)
  )
```

```
# number usage of non kakao games: treatment vs. control
# Summarize average non-Kakao game time by week and treatment group
summary_plot_data <- kakao %>%
  group_by(week, tg) %>%
  summarise(
    mean_number = mean(n_non_kakao_game, na.rm = TRUE),
    se = sd(n_non_kakao_game, na.rm = TRUE) / sqrt(n()),
    .groups = "drop"
  )
```

```
# Plot
ggplot(summary_plot_data, aes(x = week, y = mean_number,
  group = tg, color = tg, shape = tg)) +
  geom_line(size = 1) +
  geom_point(size = 3) +
  geom_errorbar(aes(ymin = mean_number - se, ymax =
mean_number + se), width = 0.1) +
  scale_shape_manual(values = c("0" = 16, "1" = 17)) + # 16 =
circle (control), 17 = triangle (treatment)
  labs(
    title = "Avg. Number of Non-Kakao Games Used, Treatment vs.
Control",
    x = "Week",
    y = "Number of Games",
    color = "Group",
    shape = "Group"
  ) +
  theme_minimal() +
  theme(
    legend.position = "right",
    legend.title = element_text(size = 10, face = "bold"),
    legend.text = element_text(size = 9)
  )
```

Appendix: R Codes

PSM / DID with 1:1 Matching, Without Replacement, Without Caliper

```
kakao_pre = subset(kakao, week == 1)
```

```
NN1 = matchit(tg ~ age + income + education + gender +
log_t_kakao_game + log_t_non_kakao_game + log_t_non_kakao +
log_t_kakao_talk + log_t_non_kakao_talk + log_t_kakao_story +
log_t_non_kakao_story + n_non_kakao + n_kakao_game +
n_non_kakao_talk + n_non_kakao_story + n_non_kakao_game,
data = kakao_pre, method = "nearest", ratio = 1)
```

```
summary(NN1)
plot(NN1, type = "hist")
```

```
#make a Data Set from Matched Sample for Follow-Up Analyses
m.dat = get_matches(NN1)
str(m.dat)
id.dat = m.dat[c("panel_id", "weights")]
m.kakao = merge(kakao, id.dat, by = "panel_id")
str(m.kakao)
```

```
## DID Model
# dummy did -- time var
did_modelNN1 <- lm(log_t_non_kakao_game ~ ii + panel_id + week +
log_t_kakao_talk + log_t_kakao_story + log_t_non_kakao_story +
log_t_kakao_game + log_t_non_kakao_talk + log_t_non_kakao +
n_non_kakao_talk + n_non_kakao_story + n_kakao_game +
n_non_kakao_game + n_non_kakao, data = m.kakao, weights =
weights)
summary(did_modelNN1)
```

```
# dummy did -- num var
did_modelNN1n <- lm(n_non_kakao_game ~ ii + panel_id + week +
log_t_kakao_talk + log_t_kakao_story + log_t_non_kakao_story +
log_t_kakao_game + log_t_non_kakao_talk + log_t_non_kakao +
n_non_kakao_talk + n_non_kakao_story + n_kakao_game +
log_t_non_kakao_game + n_non_kakao, data = m.kakao, weights =
weights)
summary(did_modelNN1n)
```

```
# FE estimation -- time var
library(plm)
panel_fe1 <- pdata.frame(m.kakao, index = c("panel_id", "week"))
```

```
fe_model1 <- plm(
log_t_non_kakao_game ~ ii + week
+ log_t_kakao_game + n_kakao_game
+ log_t_non_kakao + n_non_kakao
+ log_t_kakao_talk + log_t_kakao_story
+ log_t_non_kakao_talk + n_non_kakao_talk
+ log_t_non_kakao_story + n_non_kakao_story,
data = panel_fe1,
model = "within",
)
summary(fe_model1)
```

```
# FE estimation -- num var
fe_model1n <- plm(
n_non_kakao_game ~ ii + week
+ log_t_kakao_game + n_kakao_game
+ log_t_non_kakao + n_non_kakao
+ log_t_kakao_talk + log_t_kakao_story
+ log_t_non_kakao_talk + n_non_kakao_talk
+ log_t_non_kakao_story + n_non_kakao_story,
data = panel_fe1,
model = "within",
)
summary(fe_model1n)
```

PSM / DID with 1:1 Matching, Without Replacement, 0.05 Caliper

```
NN2 = matchit(tg ~ age + income + education + gender + log_t_kakao_game +
log_t_non_kakao_game + log_t_non_kakao + log_t_kakao_talk + log_t_non_kakao_talk
+ log_t_kakao_story + log_t_non_kakao_story + n_non_kakao + n_kakao_game +
n_non_kakao_talk + n_non_kakao_story + n_non_kakao_game, data = kakao_pre,
method = "nearest", ratio = 1, caliper = 0.05)
summary(NN2)
```

```
#make a Data Set from Matched Sample for Follow-Up Analyses
m2.dat = get_matches(NN2)
str(m2.dat)
id2.dat = m2.dat[c("panel_id", "weights")]
m2.kakao = merge(kakao, id2.dat, by = "panel_id")
str(m2.kakao)
```

```
#DID Model
#time var
did_modelINN2 <- lm(log_t_non_kakao_game ~ ii + panel_id + week +
log_t_kakao_talk + log_t_kakao_story + log_t_non_kakao_story +
log_t_kakao_game + log_t_non_kakao_talk + log_t_non_kakao +
n_non_kakao_talk + n_non_kakao_story + n_kakao_game +
n_non_kakao_game + n_non_kakao, data = m2.kakao, weights = weights)
summary(did_modelINN2)
```

```
#number var
did_modelINN2_n <- lm(n_non_kakao_game ~ ii + panel_id + week +
log_t_kakao_talk + log_t_kakao_story + log_t_non_kakao_story +
log_t_kakao_game + log_t_non_kakao_talk + log_t_non_kakao +
n_non_kakao_talk + n_non_kakao_story + n_kakao_game +
log_t_non_kakao_game + n_non_kakao, data = m2.kakao, weights =
weights)
summary(did_modelINN2_n)
```

```
# FE estimation -- time var
panel_fe2 <- pdata.frame(m2.kakao, index = c("panel_id", "week"))
```

```
fe_model2 <- plm(
log_t_non_kakao_game ~ ii + week
+ log_t_kakao_game + n_kakao_game
+ log_t_non_kakao + n_non_kakao
+ log_t_kakao_talk + log_t_kakao_story
+ log_t_non_kakao_talk + n_non_kakao_talk
+ log_t_non_kakao_story + n_non_kakao_story,
data = panel_fe2,
model = "within",
)
summary(fe_model2)
```

```
# FE estimation -- num var
fe_model2n <- plm(
n_non_kakao_game ~ ii + week
+ log_t_kakao_game + n_kakao_game
+ log_t_non_kakao + n_non_kakao
+ log_t_kakao_talk + log_t_kakao_story
+ log_t_non_kakao_talk + n_non_kakao_talk
+ log_t_non_kakao_story + n_non_kakao_story,
data = panel_fe2,
model = "within",
)
summary(fe_model2n)
```

Appendix: R Codes

PSM / DID with 1:1 Matching, Without Replacement, Caliper 0.1

```
NN3 = matchit(tg ~ age + income + education + gender +
log_t_kakao_game + log_t_non_kakao_game + log_t_non_kakao +
log_t_kakao_talk + log_t_non_kakao_talk + log_t_kakao_story +
log_t_non_kakao_story + n_non_kakao + n_kakao_game +
n_non_kakao_talk + n_non_kakao_story + n_non_kakao_game, data =
kakao_pre, method = "nearest", ratio = 1, caliper = 0.1)
summary(NN3)
```

```
#make a Data Set from Matched Sample for Follow-Up Analyses
m3.dat = get_matches(NN3)
str(m3.dat)
id3.dat = m3.dat[c("panel_id", "weights")]
m3.kakao = merge(kakao, id3.dat, by = "panel_id")
str(m3.kakao)
```

```
#DID Model
#time var
did_modelNN3 <- lm(log_t_non_kakao_game ~ ii + panel_id + week +
log_t_kakao_talk + log_t_kakao_story + log_t_non_kakao_story +
log_t_kakao_game + log_t_non_kakao_talk + log_t_non_kakao +
n_non_kakao_talk + n_non_kakao_story + n_kakao_game +
n_non_kakao_game + n_non_kakao, data = m3.kakao, weights =
weights)
summary(did_modelNN3)
```

```
#number var
did_modelNN3_n <- lm(n_non_kakao_game ~ ii + panel_id + week +
log_t_kakao_talk +
log_t_kakao_story + log_t_non_kakao_story + log_t_kakao_game
+ log_t_non_kakao_talk +
log_t_non_kakao + n_non_kakao_talk + n_non_kakao_story +
n_kakao_game +
log_t_non_kakao_game + n_non_kakao, data = m3.kakao,
weights = weights)
summary(did_modelNN3_n)
```

```
# FE estimation -- time var
panel_fe3 <- pdata.frame(m3.kakao, index = c("panel_id", "week"))
```

```
fe_model3 <- plm(
log_t_non_kakao_game ~ ii + week
+ log_t_kakao_game + n_kakao_game
+ log_t_non_kakao + n_non_kakao
+ log_t_kakao_talk + log_t_kakao_story
+ log_t_non_kakao_talk + n_non_kakao_talk
+ log_t_non_kakao_story + n_non_kakao_story,
data = panel_fe3,
model = "within",
)
summary(fe_model3)
```

```
# FE estimation -- num var
fe_model3n <- plm(
n_non_kakao_game ~ ii + week
+ log_t_kakao_game + n_kakao_game
+ log_t_non_kakao + n_non_kakao
+ log_t_kakao_talk + log_t_kakao_story
+ log_t_non_kakao_talk + n_non_kakao_talk
+ log_t_non_kakao_story + n_non_kakao_story,
data = panel_fe3,
model = "within",
)
summary(fe_model3n)
```

PSM / DID with 1:1 Matching, Without Replacement, 0.25 Caliper

```
NN4 = matchit(tg ~ age + income + education + gender + log_t_kakao_game +
log_t_non_kakao_game + log_t_non_kakao + log_t_kakao_talk + log_t_non_kakao_talk
+ log_t_kakao_story + log_t_non_kakao_story + n_non_kakao + n_kakao_game +
n_non_kakao_talk + n_non_kakao_story + n_non_kakao_game, data = kakao_pre,
method = "nearest", ratio = 1, caliper = 0.25)
summary(NN4)
plot(NN4, type = "hist")
```

```
#make a Data Set from Matched Sample for Follow-Up Analyses
m4.dat = get_matches(NN4)
str(m4.dat)
id4.dat = m4.dat[c("panel_id", "weights")]
m4.kakao = merge(kakao, id4.dat, by = "panel_id")
str(m4.kakao)
```

```
#DID Model
#time var
did_modelNN4 <- lm(log_t_non_kakao_game ~ ii + panel_id + week + log_t_kakao_talk
+ log_t_kakao_story + log_t_non_kakao_story + log_t_kakao_game +
log_t_non_kakao_talk + log_t_non_kakao + n_non_kakao_talk + n_non_kakao_story +
n_kakao_game + n_non_kakao_game + n_non_kakao, data = m4.kakao, weights =
weights)
summary(did_modelNN4)
```

```
#number var
did_modelNN4_n <- lm(n_non_kakao_game ~ ii + panel_id + week +
log_t_kakao_talk +
log_t_kakao_story + log_t_non_kakao_story + log_t_kakao_game +
log_t_non_kakao_talk +
log_t_non_kakao + n_non_kakao_talk + n_non_kakao_story +
n_kakao_game +
log_t_non_kakao_game + n_non_kakao, data = m4.kakao, weights
= weights)
summary(did_modelNN4_n)
```

```
# FE estimation -- time var
panel_fe4 <- pdata.frame(m4.kakao, index = c("panel_id", "week"))
```

```
fe_model4 <- plm(
log_t_non_kakao_game ~ ii + week
+ log_t_kakao_game + n_kakao_game
+ log_t_non_kakao + n_non_kakao
+ log_t_kakao_talk + log_t_kakao_story
+ log_t_non_kakao_talk + n_non_kakao_talk
+ log_t_non_kakao_story + n_non_kakao_story,
data = panel_fe4,
model = "within",
)
summary(fe_model4)
```

```
# FE estimation -- num var
fe_model4n <- plm(
n_non_kakao_game ~ ii + week
+ log_t_kakao_game + n_kakao_game
+ log_t_non_kakao + n_non_kakao
+ log_t_kakao_talk + log_t_kakao_story
+ log_t_non_kakao_talk + n_non_kakao_talk
+ log_t_non_kakao_story + n_non_kakao_story,
data = panel_fe4,
model = "within",
)
summary(fe_model4n)
```

Appendix: R Codes

Love Plots

```
library(cobalt)
love.plot(NN1,
  stats = "mean.diffs",
  threshold = 0.1,
  var.order = "unadjusted",
  abs = FALSE,
  colors = c("grey", "blue"),
  shapes = c("circle", "triangle"),
  sample.names = c("Before Matching", "After Matching"),
  stars = "raw",
  line = TRUE,
  title = "Standardized Mean Differences Before and After Matching")
```

```
love.plot(NN1_n,
  stats = "mean.diffs",
  threshold = 0.1,
  var.order = "unadjusted",
  abs = FALSE,
  colors = c("grey", "blue"),
  shapes = c("circle", "triangle"),
  sample.names = c("Before Matching", "After Matching"),
  stars = "raw",
  line = TRUE,
  title = "Standardized Mean Differences Before and After Matching")
```

```
love.plot(NN2,
  stats = "mean.diffs",
  threshold = 0.1,
  var.order = "unadjusted",
  abs = FALSE,
  colors = c("grey", "blue"),
  shapes = c("circle", "triangle"),
  sample.names = c("Before Matching", "After Matching"),
  stars = "raw",
  line = TRUE,
  title = "Standardized Mean Differences Before and After Matching")
```

```
love.plot(NN2_n,
  stats = "mean.diffs",
  threshold = 0.1,
  var.order = "unadjusted",
  abs = FALSE,
  colors = c("grey", "blue"),
  shapes = c("circle", "triangle"),
  sample.names = c("Before Matching", "After Matching"),
  stars = "raw",
  line = TRUE,
  title = "Standardized Mean Differences Before and After Matching")
```

```
love.plot(NN3,
  stats = "mean.diffs",
  threshold = 0.1,
  var.order = "unadjusted",
  abs = FALSE,
  colors = c("grey", "blue"),
  shapes = c("circle", "triangle"),
  sample.names = c("Before Matching", "After Matching"),
  stars = "raw",
  line = TRUE,
  title = "Standardized Mean Differences Before and After Matching")
```

```
love.plot(NN3_n,
  stats = "mean.diffs",
  threshold = 0.1,
  var.order = "unadjusted",
  abs = FALSE,
  colors = c("grey", "blue"),
  shapes = c("circle", "triangle"),
  sample.names = c("Before Matching", "After Matching"),
  stars = "raw",
  line = TRUE,
  title = "Standardized Mean Differences Before and After Matching")
```

```
love.plot(NN4,
  stats = "mean.diffs",
  threshold = 0.1,
  var.order = "unadjusted",
  abs = FALSE,
  colors = c("grey", "blue"),
  shapes = c("circle", "triangle"),
  sample.names = c("Before Matching", "After Matching"),
  stars = "raw",
  line = TRUE,
  title = "Standardized Mean Differences Before and After Matching")
```

```
love.plot(NN4_n,
  stats = "mean.diffs",
  threshold = 0.1,
  var.order = "unadjusted",
  abs = FALSE,
  colors = c("grey", "blue"),
  shapes = c("circle", "triangle"),
  sample.names = c("Before Matching", "After Matching"),
  stars = "raw",
  line = TRUE,
  title = "Standardized Mean Differences Before and After Matching")
```

```
#number var
did_modelINN4_n <- lm(n_non_kakao_game ~ ii + panel_id + week +
  log_t_kakao_talk +
    log_t_kakao_story + log_t_non_kakao_story + log_t_kakao_game +
  log_t_non_kakao_talk +
    log_t_non_kakao + n_non_kakao_talk + n_non_kakao_story +
  n_kakao_game +
    log_t_non_kakao_game + n_non_kakao, data = m4.kakao, weights
  = weights)
summary(did_modelINN4_n)
```

```
# FE estimation -- time var
panel_fe4 <- pdata.frame(m4.kakao, index = c("panel_id", "week"))
```

```
fe_model4 <- plm(
  log_t_non_kakao_game ~ ii + week
  + log_t_kakao_game + n_kakao_game
  + log_t_non_kakao + n_non_kakao
  + log_t_kakao_talk + log_t_kakao_story
  + log_t_non_kakao_talk + n_non_kakao_talk
  + log_t_non_kakao_story + n_non_kakao_story,
  data = panel_fe4,
  model = "within",
)
summary(fe_model4)
```

```
# FE estimation -- num var
fe_model4n <- plm(
  n_non_kakao_game ~ ii + week
  + log_t_kakao_game + n_kakao_game
  + log_t_non_kakao + n_non_kakao
  + log_t_kakao_talk + log_t_kakao_story
  + log_t_non_kakao_talk + n_non_kakao_talk
  + log_t_non_kakao_story + n_non_kakao_story,
  data = panel_fe4,
  model = "within",
)
summary(fe_model4n)
```

Appendix: R Codes

Love Plots of PSM 1-4

```
library(cobalt)
love.plot(NN1,
  stats = "mean.diffs",
  threshold = 0.1,
  var.order = "unadjusted",
  abs = FALSE,
  colors = c("grey", "blue"),
  shapes = c("circle", "triangle"),
  sample.names = c("Before Matching", "After Matching"),
  stars = "raw",
  line = TRUE,
  title = "Standardized Mean Differences Before and After Matching")
```

```
love.plot(NN1_n,
  stats = "mean.diffs",
  threshold = 0.1,
  var.order = "unadjusted",
  abs = FALSE,
  colors = c("grey", "blue"),
  shapes = c("circle", "triangle"),
  sample.names = c("Before Matching", "After Matching"),
  stars = "raw",
  line = TRUE,
  title = "Standardized Mean Differences Before and After Matching")
```

```
love.plot(NN2,
  stats = "mean.diffs",
  threshold = 0.1,
  var.order = "unadjusted",
  abs = FALSE,
  colors = c("grey", "blue"),
  shapes = c("circle", "triangle"),
  sample.names = c("Before Matching", "After Matching"),
  stars = "raw",
  line = TRUE,
  title = "Standardized Mean Differences Before and After Matching")
```

```
love.plot(NN2_n,
  stats = "mean.diffs",
  threshold = 0.1,
  var.order = "unadjusted",
  abs = FALSE,
  colors = c("grey", "blue"),
  shapes = c("circle", "triangle"),
  sample.names = c("Before Matching", "After Matching"),
  stars = "raw",
  line = TRUE,
  title = "Standardized Mean Differences Before and After Matching")
```

```
love.plot(NN3,
  stats = "mean.diffs",
  threshold = 0.1,
  var.order = "unadjusted",
  abs = FALSE,
  colors = c("grey", "blue"),
  shapes = c("circle", "triangle"),
  sample.names = c("Before Matching", "After Matching"),
  stars = "raw",
  line = TRUE,
  title = "Standardized Mean Differences Before and After Matching")
```

```
love.plot(NN3_n,
  stats = "mean.diffs",
  threshold = 0.1,
  var.order = "unadjusted",
  abs = FALSE,
  colors = c("grey", "blue"),
  shapes = c("circle", "triangle"),
  sample.names = c("Before Matching", "After Matching"),
  stars = "raw",
  line = TRUE,
  title = "Standardized Mean Differences Before and After Matching")
```

```
love.plot(NN4,
  stats = "mean.diffs",
  threshold = 0.1,
  var.order = "unadjusted",
  abs = FALSE,
  colors = c("grey", "blue"),
  shapes = c("circle", "triangle"),
  sample.names = c("Before Matching", "After Matching"),
  stars = "raw",
  line = TRUE,
  title = "Standardized Mean Differences Before and After Matching")
```

```
love.plot(NN4_n,
  stats = "mean.diffs",
  threshold = 0.1,
  var.order = "unadjusted",
  abs = FALSE,
  colors = c("grey", "blue"),
  shapes = c("circle", "triangle"),
  sample.names = c("Before Matching", "After Matching"),
  stars = "raw",
  line = TRUE,
  title = "Standardized Mean Differences Before and After Matching")
```

```
#number var
did_modelINN4_n <- lm(n_non_kakao_game ~ ii + panel_id + week +
  log_t_kakao_talk +
    log_t_kakao_story + log_t_non_kakao_story + log_t_kakao_game +
  log_t_non_kakao_talk +
    log_t_non_kakao + n_non_kakao_talk + n_non_kakao_story +
  n_kakao_game +
    log_t_non_kakao_game + n_non_kakao, data = m4.kakao, weights
  = weights)
summary(did_modelINN4_n)
```

```
# FE estimation -- time var
panel_fe4 <- pdata.frame(m4.kakao, index = c("panel_id", "week"))
```

```
fe_model4 <- plm(
  log_t_non_kakao_game ~ ii + week
  + log_t_kakao_game + n_kakao_game
  + log_t_non_kakao + n_non_kakao
  + log_t_kakao_talk + log_t_kakao_story
  + log_t_non_kakao_talk + n_non_kakao_talk
  + log_t_non_kakao_story + n_non_kakao_story,
  data = panel_fe4,
  model = "within",
)
summary(fe_model4)
```

```
# FE estimation -- num var
fe_model4n <- plm(
  n_non_kakao_game ~ ii + week
  + log_t_kakao_game + n_kakao_game
  + log_t_non_kakao + n_non_kakao
  + log_t_kakao_talk + log_t_kakao_story
  + log_t_non_kakao_talk + n_non_kakao_talk
  + log_t_non_kakao_story + n_non_kakao_story,
  data = panel_fe4,
  model = "within",
)
summary(fe_model4n)
```

Appendix: R Codes

PSM / DID with 2:1 Matching, With Replacement, W/O Caliper

```
NN8 = matchit(tg ~ age + income + education + gender +
log_t_kakao_game + log_t_non_kakao_game +
log_t_non_kakao + log_t_kakao_talk + log_t_non_kakao_talk +
log_t_kakao_story +
log_t_non_kakao_story + n_non_kakao + n_kakao_game +
n_non_kakao_talk +
n_non_kakao_story + n_non_kakao_game, data =
kakao_pre,
method = "nearest",
ratio = 2,
replace = TRUE)
summary(NN8)
round(mean(NN8$distance),3) #average propensity score
```

```
#Compare the Distributions of Propensity Scores between Treatment
and Control Groups, between Raw and Matched Samples
plot(NN8, type = "hist")
```

```
#make a Data Set from Matched Sample for Follow-Up Analyses
m.dat8 = get_matches(NN8)
str(m.dat8)
id.dat8 = m.dat8[c("panel_id", "weights")]
m.kakao8 = merge(kakao, id.dat8, by = "panel_id")
str(m.kakao8)
```

```
# DID Model
# dummy did -- time var
did_modelNN8 <- lm(log_t_non_kakao_game ~ ii + panel_id + week +
log_t_kakao_talk +
log_t_kakao_story + log_t_non_kakao_story +
log_t_kakao_game + log_t_non_kakao_talk +
log_t_non_kakao + n_non_kakao_game +
n_non_kakao_story + n_kakao_game +
n_non_kakao_story + n_non_kakao, data = m.kakao8,
weights = weights)
summary(did_modelNN8)
```

```
coef(summary(did_modelNN8))["ii", ] # just show treatment effect
```

```
# dummy did -- num var
did_modelNN1_n <- lm(n_non_kakao_game ~ ii + panel_id + week +
log_t_kakao_talk +
log_t_kakao_story + log_t_non_kakao_story +
log_t_kakao_game + log_t_non_kakao_talk +
log_t_non_kakao + n_non_kakao_game +
log_t_non_kakao_story + n_kakao_game +
log_t_non_kakao_story + n_non_kakao, data =
m.kakao, weights = weights)
summary(did_modelNN1_n)
```

```
coef(summary(did_modelNN1_n))["ii", ] # just show treatment
effect Matching")
```

PSM / DID with 3:1 matching, without replacement, no caliper

```
NN9 = matchit(tg ~ age + income + education + gender + log_t_kakao_game +
log_t_non_kakao_game +
log_t_non_kakao + log_t_kakao_talk + log_t_non_kakao_talk + log_t_kakao_story +
log_t_non_kakao_story + n_non_kakao + n_kakao_game + n_non_kakao_talk +
n_non_kakao_story + n_non_kakao_game, data = kakao_pre,
method = "nearest",
ratio = 3,
replace = FALSE)
summary(NN9)
round(mean(NN9$distance),3) #average propensity score
```

```
#Compare the Distributions of Propensity Scores between Treatment and Control Groups,
between Raw and Matched Samples
plot(NN9, type = "hist")
```

```
#make a Data Set from Matched Sample for Follow-Up Analyses
m.dat9 = get_matches(NN9)
str(m.dat9)
id.dat9 = m.dat9[c("panel_id", "weights")]
m.kakao9 = merge(kakao, id.dat9, by = "panel_id")
str(m.kakao9)
```

```
## DID Model
# dummy did -- time var
did_modelNN9 <- lm(log_t_non_kakao_game ~ ii + panel_id + week + log_t_kakao_talk +
log_t_kakao_story + log_t_non_kakao_story + log_t_kakao_game + log_t_non_kakao_talk +
log_t_non_kakao + n_non_kakao_talk + n_non_kakao_story + n_kakao_game + n_non_kakao_game +
n_non_kakao, data = m.kakao9, weights = weights)
summary(did_modelNN9)
```

```
# dummy did -- num var
did_modelNN9_n <- lm(n_non_kakao_game ~ ii + panel_id + week + log_t_kakao_talk +
log_t_kakao_story + log_t_non_kakao_story + log_t_kakao_game + log_t_non_kakao_talk +
log_t_non_kakao + n_non_kakao_talk + n_non_kakao_story + n_kakao_game +
log_t_non_kakao_story + n_non_kakao, data = m.kakao9, weights = weights)
summary(did_modelNN9_n)
```

```
# FE estimation -- time var
library(plm)
panel_fe9 <- pdata.frame(m.kakao9, index = c("panel_id", "week"))
```

```
fe_model9 <- plm(log_t_non_kakao_game ~ ii + week + log_t_kakao_game + n_kakao_game +
log_t_non_kakao + n_non_kakao + log_t_kakao_talk + log_t_kakao_story + log_t_non_kakao_talk +
n_non_kakao_talk +
log_t_non_kakao_story + n_non_kakao_story, data = panel_fe9, model = "within",
summary(fe_model9)
```

```
# FE estimation -- num var fe_model9n <- plm(n_non_kakao_game ~ ii + week +
log_t_kakao_game + n_kakao_game + log_t_non_kakao + n_non_kakao + log_t_kakao_talk +
log_t_kakao_story + log_t_non_kakao_talk + n_non_kakao_talk + log_t_non_kakao_story +
n_non_kakao_story, data = panel_fe9, model = "within",
summary(fe_model9n)
```

PSM / DID with 3:1 Matching, W/O Replacement, Caliper = 0.05

```
NN10 = matchit(tg ~ age + income + education + gender + log_t_kakao_game +
log_t_non_kakao_game +
log_t_non_kakao + log_t_kakao_talk + log_t_non_kakao_talk + log_t_kakao_story +
log_t_non_kakao_story + n_non_kakao + n_kakao_game + n_non_kakao_talk +
n_non_kakao_story + n_non_kakao_game, data = kakao_pre,
method = "nearest",
ratio = 3,
replace = FALSE,
caliper = 0.05)
summary(NN10)
round(mean(NN10$distance),3) #average propensity score
```

```
#Compare the Distributions of Propensity Scores between Treatment and Control Groups,
between Raw and Matched Samples
plot(NN10, type = "hist")
```

```
#make a Data Set from Matched Sample for Follow-Up Analyses
m.dat10 = get_matches(NN10)
str(m.dat10)
```

```
id.dat10 = m.dat10[c("panel_id", "weights")]
```

```
m.kakao10 = merge(kakao, id.dat10, by = "panel_id")
str(m.kakao10)
```

```
## DID Model
# dummy did -- time var
did_modelNN10 <- lm(log_t_non_kakao_game ~ ii + panel_id + week + log_t_kakao_talk +
log_t_kakao_story + log_t_non_kakao_story + log_t_kakao_game +
log_t_non_kakao_talk +
log_t_non_kakao + n_non_kakao_talk + n_non_kakao_story + n_kakao_game +
n_non_kakao_game + n_non_kakao, data = m.kakao10, weights = weights)
summary(did_modelNN10)
```

```
coef(summary(did_modelNN10))["ii", ] # just show treatment effect
```

```
# dummy did -- num var
did_modelNN10_n <- lm(n_non_kakao_game ~ ii + panel_id + week + log_t_kakao_talk +
log_t_kakao_story + log_t_non_kakao_story + log_t_kakao_game +
log_t_non_kakao_talk +
log_t_non_kakao + n_non_kakao_talk + n_non_kakao_story + n_kakao_game +
log_t_non_kakao_story + n_non_kakao, data = m.kakao10, weights = weights)
summary(did_modelNN10_n)
```

```
coef(summary(did_modelNN10_n))["ii", ] # just show treatment effect
```

Appendix: R Codes

```
# FE estimation -- time var
panel_fe10 <- pdata.frame(m.kakao10, index = c("panel_id", "week"))
```

```
fe_model10 <- plm(
  log_t_non_kakao_game ~ ii + week
  + log_t_kakao_game + n_kakao_game
  + log_t_non_kakao + n_non_kakao
  + log_t_kakao_talk + log_t_kakao_story
  + log_t_non_kakao_talk + n_non_kakao_talk
  + log_t_non_kakao_story + n_non_kakao_story,
  data = panel_fe10,
  model = "within",
)
summary(fe_model10)
```

```
# FE estimation -- num var
fe_model10n <- plm(
  n_non_kakao_game ~ ii + week
  + log_t_kakao_game + n_kakao_game
  + log_t_non_kakao + n_non_kakao
  + log_t_kakao_talk + log_t_kakao_story
  + log_t_non_kakao_talk + n_non_kakao_talk
  + log_t_non_kakao_story + n_non_kakao_story,
  data = panel_fe10,
  model = "within",
)
summary(fe_model10n)
```

Love Plots

```
library(cobalt)
love.plot(NN8,
  stats = "mean.diffs",
  threshold = 0.1,
  var.order = "unadjusted",
  abs = FALSE,
  colors = c("grey", "blue"),
  shapes = c("circle", "triangle"),
  sample.names = c("Before Matching", "After Matching"),
  stars = "raw",
  line = TRUE,
  title = "Standardized Mean Differences Before and After Matching")
```

```
love.plot(NN9,
  stats = "mean.diffs",
  threshold = 0.1,
  var.order = "unadjusted",
  abs = FALSE,
  colors = c("grey", "blue"),
  shapes = c("circle", "triangle"),
  sample.names = c("Before Matching", "After Matching"),
  stars = "raw",
  line = TRUE,
  title = "Standardized Mean Differences Before and After Matching")
```

```
love.plot(NN10,
  stats = "mean.diffs",
  threshold = 0.1,
  var.order = "unadjusted",
  abs = FALSE,
  colors = c("grey", "blue"),
  shapes = c("circle", "triangle"),
  sample.names = c("Before Matching", "After Matching"),
  stars = "raw",
  line = TRUE,
  title = "Standardized Mean Differences Before and After Matching")
```

PSM with 2:1 Matching, without Replacement, W/O Caliper

```
NN2.out <- matchit(
  formula = tg ~ age + income + education + gender +
  log_t_kakao_game + n_kakao_game +
  log_t_non_kakao_game + n_non_kakao_game +
  log_t_kakao_talk + log_t_kakao_story +
  log_t_non_kakao_talk + n_non_kakao_talk +
  log_t_non_kakao_story + n_non_kakao_story,
  data = kakao_pre,
  method = "nearest",
  ratio = 2,
  replace = FALSE
)
```

```
panel_matched <- panel_matched %>%
  mutate(
    week = factor(week, levels = c(1, 2)),
    income = factor(income),
    education = factor(education),
    gender = factor(gender)
  )
model_dummy <- lm(log_t_non_kakao_game ~ ii + panel_id + week + age + income +
  education + gender + log_t_kakao_talk +
  log_t_kakao_story + log_t_non_kakao_story + log_t_kakao_game +
  log_t_non_kakao_talk +
  log_t_non_kakao + n_non_kakao_talk + n_non_kakao_story + n_kakao_game +
  n_non_kakao, data = panel_matched, weights = weights)
summary(model_dummy)
```

```
library(plm)
```

```
panel_fe <- pdata.frame(panel_matched, index = c("panel_id", "week"))
```

```
fe_model <- plm(
  log_t_non_kakao_game ~ ii + week
  + log_t_kakao_game + n_kakao_game
  + log_t_non_kakao + n_non_kakao
  + log_t_kakao_talk + log_t_kakao_story
  + log_t_non_kakao_talk + n_non_kakao_talk
  + log_t_non_kakao_story + n_non_kakao_story,
  data = panel_fe,
  model = "within",
)
```

```
summary(fe_model)
```

```
NN2.out2 <- matchit(
  formula = tg ~ age + income + education + gender +
  log_t_kakao_game + n_kakao_game +
  log_t_non_kakao_game + n_non_kakao_game +
  log_t_kakao_talk + log_t_kakao_story +
  log_t_non_kakao_talk + n_non_kakao_talk +
  log_t_non_kakao_story + n_non_kakao_story,
  data = kakao_pre,
  method = "nearest",
  caliper = 0.25,
  ratio = 1,
  replace = FALSE
)
```

```
model_dummy <- lm(log_t_non_kakao_game ~ ii + panel_id + week + age + income +
  education + gender + log_t_kakao_talk +
  log_t_kakao_story + log_t_non_kakao_story + log_t_kakao_game +
  log_t_non_kakao_talk +
  log_t_non_kakao + n_non_kakao_talk + n_non_kakao_story + n_kakao_game +
  n_non_kakao, data = panel_matched, weights = weights)
```

```
summary(model_dummy)
```

```
library(plm)
```

```
panel_fe <- pdata.frame(panel_matched, index = c("panel_id", "week"))
```

```
fe_model <- plm(
  log_t_non_kakao_game ~ ii + week
  + log_t_kakao_game + n_kakao_game
  + log_t_non_kakao + n_non_kakao
  + log_t_kakao_talk + log_t_kakao_story
  + log_t_non_kakao_talk + n_non_kakao_talk
  + log_t_non_kakao_story + n_non_kakao_story,
  data = panel_fe,
  model = "within",
)
```

```
summary(fe_model)
```

```
NN2.out3 <- matchit(
  formula = tg ~ age + income + education + gender +
  log_t_kakao_game + n_kakao_game +
  log_t_non_kakao_game + n_non_kakao_game +
  log_t_non_kakao + n_non_kakao +
  log_t_kakao_talk + log_t_kakao_story +
  log_t_non_kakao_talk + n_non_kakao_talk +
  log_t_non_kakao_story + n_non_kakao_story,
  data = kakao_pre,
  method = "nearest",
  caliper = 0.05,
  ratio = 2,
  replace = FALSE
)
model_dummy <- lm(log_t_non_kakao_game ~ ii + panel_id + week + age + income +
education + gender + log_t_kakao_talk +
  log_t_kakao_story + log_t_non_kakao_story + log_t_kakao_game +
log_t_non_kakao_talk +
  log_t_non_kakao + n_non_kakao_talk + n_non_kakao_story + n_kakao_game +
n_non_kakao_game + n_non_kakao, data = panel_matched, weights = weights)
```

```
summary(model_dummy)
```

```
panel_fe <- pdata.frame(panel_matched, index = c("panel_id", "week"))
```

```
fe_model <- plm(
  log_t_non_kakao_game ~ ii + week
  + log_t_kakao_game + n_kakao_game
  + log_t_non_kakao + n_non_kakao
  + log_t_kakao_talk + log_t_kakao_story
  + log_t_non_kakao_talk + n_non_kakao_talk
  + log_t_non_kakao_story + n_non_kakao_story,
  data = panel_fe,
  model = "within",
)
```

```
summary(fe_model)
```

Number of rival games

```
NN2.out <- matchit(
  formula = tg ~ age + income + education + gender +
  log_t_kakao_game + n_kakao_game +
  log_t_non_kakao_game + n_non_kakao_game +
  log_t_non_kakao + n_non_kakao +
  log_t_kakao_talk + log_t_kakao_story +
  log_t_non_kakao_talk + n_non_kakao_talk +
  log_t_non_kakao_story + n_non_kakao_story,
  data = kakao_pre,
  method = "nearest",
  ratio = 2,
  replace = FALSE
)
```

```
model_dummy <- lm(n_non_kakao_game ~ ii + panel_id + week + age + income + education +
gender + log_t_kakao_talk +
  log_t_kakao_story + log_t_non_kakao_story + log_t_kakao_game +
log_t_non_kakao_talk +
  log_t_non_kakao + n_non_kakao_talk + n_non_kakao_story + n_kakao_game
+ n_non_kakao, data = panel_matched, weights = weights)
summary(model_dummy)
```

```
panel_fe <- pdata.frame(panel_matched, index = c("panel_id", "week"))
```

```
fe_model <- plm(
  n_non_kakao_game ~ ii + week
  + log_t_kakao_game + n_kakao_game
  + log_t_non_kakao + n_non_kakao
  + log_t_kakao_talk + log_t_kakao_story
  + log_t_non_kakao_talk + n_non_kakao_talk
  + log_t_non_kakao_story + n_non_kakao_story,
  data = panel_fe,
  model = "within",
)
```

```
summary(fe_model)
```

```
NN2.out2 <- matchit(
  formula = tg ~ age + income + education + gender +
  log_t_kakao_game + n_kakao_game +
  log_t_non_kakao_game + n_non_kakao_game +
  log_t_non_kakao + n_non_kakao +
  log_t_kakao_talk + log_t_kakao_story +
  log_t_non_kakao_talk + n_non_kakao_talk +
  log_t_non_kakao_story + n_non_kakao_story,
  data = kakao_pre,
  method = "nearest",
  caliper = 0.25,
  ratio = 1,
  replace = FALSE
)
```

```
model_dummy <- lm(n_non_kakao_game ~ ii + panel_id + week + age + income + education +
gender + log_t_kakao_talk +
  log_t_kakao_story + log_t_non_kakao_story + log_t_kakao_game +
log_t_non_kakao_talk +
  log_t_non_kakao + n_non_kakao_talk + n_non_kakao_story + n_kakao_game +
n_non_kakao, data = panel_matched, weights = weights)
```

```
summary(model_dummy)
```

```
panel_fe <- pdata.frame(panel_matched, index = c("panel_id", "week"))
```

```
fe_model <- plm(
  n_non_kakao_game ~ ii + week
  + log_t_kakao_game + n_kakao_game
  + log_t_non_kakao + n_non_kakao
  + log_t_kakao_talk + log_t_kakao_story
  + log_t_non_kakao_talk + n_non_kakao_talk
  + log_t_non_kakao_story + n_non_kakao_story,
  data = panel_fe,
  model = "within",
)
```

```
summary(fe_model)
```

```
NN2.out3 <- matchit(
  formula = tg ~ age + income + education + gender +
  log_t_kakao_game + n_kakao_game +
  log_t_non_kakao_game + n_non_kakao_game +
  log_t_non_kakao + n_non_kakao +
  log_t_kakao_talk + log_t_kakao_story +
  log_t_non_kakao_talk + n_non_kakao_talk +
  log_t_non_kakao_story + n_non_kakao_story,
  data = kakao_pre,
  method = "nearest",
  caliper = 0.05,
  ratio = 2,
  replace = FALSE
)
```

```
model_dummy <- lm(n_non_kakao_game ~ ii + panel_id + week + age + income + education +
gender + log_t_kakao_talk +
  log_t_kakao_story + log_t_non_kakao_story + log_t_kakao_game +
log_t_non_kakao_talk +
  log_t_non_kakao + n_non_kakao_talk + n_non_kakao_story + n_kakao_game +
n_non_kakao, data = panel_matched, weights = weights)
```

```
summary(model_dummy)
```

```
panel_fe <- pdata.frame(panel_matched, index = c("panel_id", "week"))
```

```
fe_model <- plm(
  n_non_kakao_game ~ ii + week
  + log_t_kakao_game + n_kakao_game
  + log_t_non_kakao + n_non_kakao
  + log_t_kakao_talk + log_t_kakao_story
  + log_t_non_kakao_talk + n_non_kakao_talk
  + log_t_non_kakao_story + n_non_kakao_story,
  data = panel_fe,
  model = "within",
)
```

```
summary(fe_model)
```